MIL-C-46909D(EA)
12 May 1987
SUPERSEDING
MIL-C-46909C(EA)
5 December 1967

MILITARY SPECIFICATION

CHARGE, EXPELLING, FOR PROJECTILE, HC SMOKE, 155 MM, M116A1

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of expelling charge to be used in the 155 mm M116A1 HC Smoke Projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-117 - Bags, Sleeves, and Tubing-Interior Packaging.
MIL-P-223 - Powder, Black.
MIL-B-2427 - Box, Ammunition Packing: Wood, Nailed.
MIL-B-3106 - Board, Composition, Water-Resistant, Solid (For Filler or Cushioning Pads).
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1234 - Pyrotechnics: Sampling, Inspection and Testing.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

7549275 - Pallet for 155 MM Projectile.
8796522 - Marking Diagram and Sealing for Wood Packing Boxes.
9352713 - Expelling Charge Assembly.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

8885153 - Plate, Baffle.
9255386 - Expelling Charge Support.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The expelling charge cup and cover shall be molded using the high density, stress crack resistant grade polyethylene specified on the applicable drawings.

3.2 Expelling charge assembly. The expelling charge assembly shall comply with all requirements specified on Drawing 9352713 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder at the loading station shall not exceed 0.7 percent, when tested as specified in 4.6.1.

3.4 Leak test of expelling charge assemblies. Expelling charge assemblies shall not leak when tested as specified in 4.6.2.

3.5 Compressive load test. Expelling charge assemblies shall not leak when tested as specified in 4.6.3.

3.6 Workmanship. The expelling charge components and assembly shall be free from cracks, air bubbles, holes, blisters, chips, dirt, grease and foreign matter. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.7 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Quality assurance terms and definitions. Refer to MIL-STD-109 for definitions of quality assurance terms.

4.2 Classification of inspections.

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contract for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 10 of each individual component from each mold cavity which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including unit packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submisison, the contractor shall inspect the same to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contractng officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. The contractor shall submit a welding procedure, to include the type of welder, model number, size and welding cycle information to the Government for approval prior to proceeding with regular production.

4.3.2 Inspections to be performed. Components will be subjected by the Government to any or all of the examinations and tests specified in 4.4.3 and 4.5 of this specification and any or all requirements of the applicable drawings.

4.3.3 Rejection. If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to termnate its inspection upon any failure of a component to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Lot formation. See MIL-A-48078.

4.4.2 Submission of product. The product shall be submitted in accordance with MIL-STD-105. Unless otherwise specified, an AQL of 0.40 percent defective for each major defect and an AQL of 0.65 percent defective for each minor defect shall be used.

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4.4.3 Classification of defects.

4.4.3.1 Cup (Drawing 9352714).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Outside diameter of flange incorrect	Gage
102	Outside diameter under flange incorrect	Gage
103	Thickness of flange incorrect	Gage
104	Outside diameter of closed end incorrect	Gage
105	Wall thickness incorrect	Gage
106	Wall thickness of recess in closed end incorrect	Gage
107	Diameter of recess incorrect	Gage
108	Overall height incorrect	Gage
109	Flashing in recess area	Gage
110	Radius improper at intersection of outside wall and flange underside	Gage
111	Height of energy director ridge incorrect	Gage
112	Location of energy director ridge incorrect	Gage
113	Inside diameter of flange incorrect	Gage
114	Evidence of poor workmanship (see 3.6)	Visual
115	Material incorrect (100 percent)	Certification by test (furnish test results)
116	Mold marks, ribs, fins or flashings incorrect	Gage

4.4.3.2 Cover (Drawing 9352715).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Thickness incorrect	Gage
102	Major diameter incorrect	Gage
103	Diameter of wiping edge incorrect	Gage
104	Location of welding groove incorrect	Gage
105	Evidence of poor workmanship (see 3.6)	Visual
106	Height of snap closure diameter incorrect	Gage
107	Material incorrect (100 percent)	Certification by test (furnish test results)

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4.4.3.3 Tool control. In lieu of performing sampling inspections listed under classification of defects on each lot, the following tool control evaluations may be substituted. Dimensionally examine (100 percent) 5 each expelling charge bodies and covers from each mold cavity for conformance to all dimensions on Drawings 9352714 and 9352715 to qualify the mold cavity. To maintain mold cavity qualification, one body and one cover from each mold cavity shall be examined 100 percent for conformance to Drawings 9352714 and 9352715 for each week's production or when a mold cavity is repaired or changed. In the event a mold is found defective, all bodies and covers from the defective mold cavity produced since the last acceptable body and cover examination shall be sampled using MIL-STD-105, AQL 0.40 percent. If the quantity is rejected, the bodies and covers shall be screened and defective bodies and covers removed prior to using for production. Each cavity shall be identified by a number.

4.4.3.4 Expelling charge assembly, prior to sealing (Drawing 9352713).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 4.0 percent defective	
101	Powder charge missing	Visual
102	Powder charge weight incorrect	Scale/volume
103	Foreign matter in cup or on welding surfaces	Visual
104	Evidence of poor workmanship (see 3.6)	Visual

4.4.3.5 Expelling charge assembly, sealed (Drawing 9352713).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Assembly fails leak test	Visual
102	Black powder between flange of cup and cover	Visual
	AQL 0.40 percent defective	
103	Outside diameter of flange incorrect	Gage
104	Height incorrect	Gage
105	Flange thickness incorrect	Gage
Minor:	AQL 0.65 percent defective	
201	Container damaged	Visual
202	Marking incorrect (Note 5)	Visual

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4.4.3.6 Unsealed fiber container (see 5.1.2.1 and 5.2.1).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Any packing component missing or improperly assembled	Visual
Minor:	AQL 0.65 percent defective	
201	Expelling charge assembly missing	Visual

4.4.3.7 Sealed fiber container (see 5.1.2.1 and 5.2.1).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Container damaged	Visual
102	Tape incomplete, improperly positioned, or badly wrinkled	Visual
Minor:	AQL 0.65 percent defective	
201	Contents loose	Manual

4.4.3.8 Unsealed packing box (see 5.2).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Any packing component missing or improperly assembled	Visual

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4.4.3.9 Sealed packing box (see 5.2 and 5.3).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.40 percent defective	
101	Box damaged exposing contents	Visual
102	DOT marking missing or incorrect	Visual
103	Board broken	Visual
104	Strapping missing, broken or improperly applied (when applicable)	Visual
Minor:	AQL 0.65 percent defective	
201	Marking missing or incorrect	Visual
202	Contents loose	Manual

4.5 Inspection procedure for tests.

4.5.1 Moisture content of black powder. The moisture and volatile content shall be determined in accordance with 4.6.1. One sample of black powder from the expelling charge shall be examined from each 8 hours production. If the sample moisture content fails to meet the specified requirements (see 3.3), the 8 hours production represented by the sample shall be rejected.

4.5.2 Leak test of expelling charge assemblies. Leak testing of the expelling charge assemblies shall be in accordance with 4.6.2. Reject all assemblies which leak.

4.5.3 Compressive load test. From each hour's production the last two expelling charge assemblies which have passed the leak test (see 4.6.2) shall be subjected to the compressive load test in accordance with 4.6.3 and then leak tested again in accordance with 4.6.2. If the sample fails to comply with the leak test requirement, the hour production represented by the sample shall be rejected.

4.6 Tests. Tests shall be conducted as follows:

4.6.1 Moisture content of black powder. The moisture content of the black powder shall be determined in accordance with method 101.1, 102.1, 102.2 of MIL-STD-1234 (see 6.3). If the moisture content of the powder tested exceeds 0.7 percent, the lot of charge assemblies shall be rejected.

4.6.2 Leak test. Expelling charge assemblies shall be leak tested in accordance with test method 118, paragraph 5.2 of MIL-STD-331.

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4.6.3 Compressive load testing. Place the selected expelling charge assemblies through the center of an expelling charge support (Drawing 9255386) so that the underside of the expelling charge assembly rests on the support. Take an aluminum cylinder 3.0 -0.1 inch inside diameter, 0.20 to 0.10 inch wall; 2.5 ± 0.5 inches long and stand on end. Place the expelling charge assembly in the support and on the aluminum cylinder so that the bottom of the cup extends into the cylinder and the support rests on the cylinder end. Place an expelling charge baffle plate (Drawing 8885153) on top of the expelling charge assembly. Apply a 200 pound load to the baffle plate for a minimum period of 1.0 minute. Remove the load and leak test the expelling charge assembly as specified in 4.6.2.

5. PACKAGING

5.1 Unit packing.

5.1.1 Unit packing of plastic components. Expelling charge cups (Drawing 9352714) and covers (Drawing 9352715) shall be unit packed in accordance with commercial practice adequate to insure acceptance and safe delivery by the carrier for the mode of transportation employed. This unit packing shall include nesting of the cups and covers.

5.1.2 Unit packing of expelling charge assemblies.

5.1.2.1 Level A. Fifty M116A1 expelling charges shall be unit packed in a PPP-B-636, style CSSC, class domestic, grade 200 fiberboard box complete with filler pads and separators consisting of two half slotted partitions assembled to form 50 individual cells in a 5 by 5 by 2 layer per box arrangement. A filler pad shall be placed between each layer of charges, and one or more filler pads shall be added to the top layer of charges as required to insure a tight pack. Separators and filler pads shall be constructed of fiberboard conforming to class domestic, variety SW, grade 200 of PPP-F-320. The box size shall be approximately 15-3/4 by 15-3/4 by 3-3/4 inches deep (inside dimensions). Separator cell size shall be approximately 15-5/8 by 15-5/8 inches. The boxes shall be closed by applying 2 inch wide tape meeting requirements for grade B, type III of PPP-T-45 over the entire length of the top and bottom center seam of the box with an approximate 2 inch extension of tape onto the end panels of the box. Each fiberboard box shall be inserted into a barrier bag conforming to MIL-B-117, type I, class E. All excess air shall be exhausted from the bag and the bag closed by heat sealing. The bag closure shall meet the heat-seal test contained in MIL-P-116. Bag size shall be approximately 21 by 22-1/2 inches.

5.1.2.2 Level C. Expelling charges shall be unit packed as specified in 5.1.1 or 5.1.2. Barrier bag shall not be required.

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5.2 Packing.

5.2.1 Level A. Two fiberboard boxes (100 M116A1 charges) shall be packed in a wood ammunition box conforming to MIL-B-2427, grade A, type I, class 1. Fillers of composition board, MIL-B-3106, type I or II shall be inserted at the sides, ends of the box, and on top of the contents as required to insure a tight pack. The box size shall be approximately 16-3/8 by 16-1/8 by 8-3/4 inches deep (inside dimensions). Side and end filler shall be approximately 16 by 8 by 1/16 inches thick. Top fillers shall be approximately 16 by 16 by 1/16 inches thick.

5.2.2 Level C. Expelling charges shall be packed as specified in 5.2.1 except boxes shall be grade C, class 1 or 2 (hardware will simplify reuse of wood boxes) and fillers shall be class domestic, variety SW, grade 200 of PPP-F-320.

5.3 Marking.

5.3.1 Level A. Each fiberboard box and barrier bag shall be marked with black water-proof ink as follows:

(QYT) CHARGE, EXPELLING, FOR PROJECTILE, 155 MM, (M116A1)

LOT NUMBER PACKED (MO/YR)

Letters shall all be capitals. Height of all marking shall be minimum 1/4 inch. Boxes and barrier bags shall be marked by stenciling, stamping, or printing. Exterior wood box shall be marked in accordance with Drawing 8796522. The Department of Transportation (DOT) marking shall be "BLACK POWDER". The words "THIS SIDE UP" shall appear on the top of the box in minimum 1 inch letters.

5.3.2 Level C. Shipping container shall be marked in accordance with Drawing 8796522. DOT and precuationary marking specified in 5.3.1 shall apply.

6. NOTES

6.1 Intended use. This item is intended for use in the 155 MM M116A1 HC smoke projectile.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Ammunition data cards should be prepared for each lot in accordance with MIL-STD-1167.

6.3 Black powder. Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal from an expelling charge assembly, and to run the determination as soon as possible thereafter.

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6.4 Subject term (key word) listing.

Black powder
Charge, expelling
Projectile, HC smoke

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-46909D(EA)	CHARGE, EXPELLING, FOR PROJECTILE, HC SMOKE, 155 MM, M116A1

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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